SU(2) Wilson Lattice Gauge Structure on the GeoVac Hopf Graph: The Non-Abelian Sibling of Paper 25

GeoVac Project

April 2026

Abstract

Paper 25 [18] placed the GeoVac Hopf graph inside the Wilson–Hodge lattice-gauge vocabulary with a $U(1)$ connection on edges, and explicitly left as open (§VII.A) the question of whether a non-abelian gauge structure is natural on the same graph. This paper answers that question for the compact simply-connected non-abelian Lie group of rank 1, SU(2). The coincidence $S^3 = \mathrm{SU}(2)$ (as manifolds) makes this case structurally special: the underlying manifold of the GeoVac graph *is* the gauge group. Taking link variables $U_e \in \mathrm{SU}(2)$, plaquettes as the primitive closed non-backtracking walks of the Ihara zeta setup [20], and the Wilson action $S_W[U] = \beta \sum_P \big(1 - \frac{1}{2}\mathrm{Re}\,\mathrm{tr}\,U_P\big)$, we obtain a gauge-invariant finite-dimensional lattice gauge theory with Haar-normalizable partition function. Three structural results follow. (1) *$U(1)$ reduction to Paper 25.* The diagonal (maximal-torus) sector of SU(2) link variables reproduces Paper 25's $U(1)$ Wilson–Hodge action to machine precision. Paper 25 is the Cartan-subalgebra projection of the present theory; the abelian construction it documents is a structural sub-sector of a genuinely non-abelian one. (2) *The weak-coupling kinetic term completes the Hodge-1 Laplacian.* Linearizing S_W around the trivial vacuum $U_e = I$ produces, at second order in the gauge field, the plaquette (curl) quadratic form $(\beta/8)\sum_a \langle A^a, B_2 B_2^\top A^a\rangle$ on $\mathfrak{su}(2)$-valued 1-cochains, with per-plaquette per-component coefficient $1/8 = 1/(4N_c)$. This curl form is *not* a scalar multiple of Paper 25's edge Laplacian $L_1 = B^\top B$ (the two have complementary supports: the plaquette columns of B_2 span $\ker L_1$ exactly); it is the co-exact part that completes L_1 to the full discrete Hodge-1 Laplacian $B^\top B + B_2 B_2^\top$ of the 2-complex with the Ihara plaquettes attached as faces, exactly the face structure Paper 25 §II anticipates. (3) *Plaquette enumeration and Monte Carlo Wilson-loop data.* On the S^3 Coulomb graph at $n_{\max} = 2, 3, 4$, the first-Betti counts are $\beta_1 = 0, 2, 8$ and the primitive 4-cycle counts are $0, 2, 8$; a 2000-sample Metropolis run on $n_{\max} = 3$ at $\beta \in \{0.5, 1, 2, 5\}$ gives Wilson expectations $\langle W\rangle = 0.126, 0.158, 0.332, 0.681$, monotonic with β and matched to the leading-order character expansion at the 2–35% level across the four β values (closest at the strong- and weak-coupling ends). All 36 tests in `tests/test_su2_wilson_gauge.py` pass. The scope is a compact-graph non-abelian lattice gauge construction, not a Yang–Mills mass-gap claim on $\mathbb{R}^4$; the paper closes with a synthesis observation that the GeoVac framework now carries *five* action principles—maximum entropy on nodes, Rayleigh–Ritz on node-wavefunctions, Wilson action on gauge edges, Ihara path-integral on walks, and spectral action on Dirac spectrum—suggesting—though we do not claim—a framework-wide organizing principle of least action. An SU(3) sibling on the Bargmann–Segal S^5 graph (Paper 24) has been constructed in Sprint ST-SU3 (`debug/st_su3_wilson_memo.md`, May 2026), confirming Proposition 1's analog (Cartan reduction to $U(1) \times U(1)$) and Proposition 2's analog (kinetic coefficient $1/12 = 1/(4N_c)$) at machine precision; matter coupling on S^5 encounters the Sprint 5 Track S5 Clebsch–Gordan-intertwiner obstruction, sharpening Paper 24's Coulomb/HO asymmetry to four layers (spectrum-computing L_0, calibration π, Wilson gauge with natural matter coupling, and the universal Wilson–Hodge vocabulary; the first three Coulomb-specific, the fourth universal).

1 Introduction

The GeoVac framework discretizes Coulomb bound-state quantum mechanics onto a finite graph that converges to the unit three-sphere S^3 in the continuum limit [13]. Paper 25 [18] observed that the same graph, read through its signed node–edge incidence matrix B, carries all the combinatorial ingredients of a lattice gauge theory: nodes as matter sites, edges as link variables in a $U(1)$ connection, node Laplacian $L_0 = BB^\top$ as matter propagator, edge Laplacian $L_1 = B^\top B$ as discrete Hodge-1 Laplacian. Section VII.A of Paper 25 explicitly flagged as open the question of whether a non-abelian gauge structure is also natural on the same graph, and sharpened the question by an analysis of the S^5 Bargmann–Segal graph of Paper 24 [17] which *fails* to admit an SU(3) lattice gauge analog—the $(N, 0)$ representation tower has Clebsch–Gordan intertwiners on links, not fixed group elements, so it fails the Wilson fixed-group-on-every-link requirement.

The S^3 case is categorically different. As manifolds, $S^3 = \mathrm{SU}(2)$: the underlying manifold of the GeoVac graph *is* the gauge group. Taking SU(2)-valued link variables is therefore not a choice made by matching representation content across shells; it is the natural object carried by the manifold itself. This is the cleanest geometric setting in which a non-abelian Wilson lattice gauge theory can be built on a GeoVac graph.

This paper reports the construction. The mathematics is standard [1, 2][1]: link variables $U_e \in \mathrm{SU}(2)$, plaquette holonomies $U_P = \prod_{e\in P} U_e$, Wilson action $S_W = \beta \sum_P (1 - \frac{1}{2}\mathrm{Re}\,\mathrm{tr}\,U_P)$, node-local gauge transformations $U_e \mapsto g_{s(e)} U_e\, g^\dagger_{t(e)}$, and Haar measure on each link. What is new is (i) the choice of plaquettes as the primitive closed non-backtracking walks of Paper 29 [20]'s Ihara zeta machinery, rather than the elementary four-cycles of a hypercubic lattice, (ii) the confirmation that the maximal-torus reduction is *exactly* Paper 25's $U(1)$ construction, and (iii) the identification of the weak-coupling kinetic term as the plaquette (curl) quadratic form that completes Paper 25's edge Laplacian L_1 to the full discrete Hodge-1 Laplacian of the plaquette-filled 2-complex.

Scope. This is an observation paper. What it establishes is a self-consistent non-abelian lattice gauge structure on the finite compact GeoVac Hopf graph, together with three structural results. What it does *not* establish is a continuum-limit Yang–Mills mass gap on $\mathbb{R}^4$, confinement (the graph has a single plaquette class at the sizes tested, making the area/perimeter-law diagnostic inapplicable), or any new derivation of α in the Paper 2 sense. The rhetoric follows the CLAUDE.md §1.5 rule: the mathematical construction is sharp; the physical interpretation is modest. The construction is a non-abelian instance of the Marcolli–van Suijlekom gauge-network framework [7], which defines finite spectral triples on graphs whose spectral action is a Wilson lattice gauge theory; Perez-Sanchez [8] establishes that the continuum limit in that lineage is Yang–Mills without Higgs, consistent with the scope statements of Section 7.

Organization. Section 2 recaps the Paper 25 Wilson–Hodge vocabulary in the form needed. Section 3 states the SU(2) link variables, plaquette structure, Wilson action, and gauge transformation. Section 4 gives the character expansion of the partition function on SU(2). Section 5 reports the three structural results: $U(1)$-reduction exactness, the kinetic-term / Hodge-completion result, and plaquette/Wilson-loop data. Section 6 places the construction inside a framework-wide synthesis: the GeoVac project now exhibits five independent action principles, each organizing a different

[1] Concurrent work in the Bonn/DESY Hamiltonian-lattice program develops SU(2) Hamiltonian lattice gauge theory using discrete partitionings of the gauge group, with motivation toward quantum simulation: see Garofalo et al. [9] for the SU(2) Hamiltonian formulation based on S_3 partitionings (here S_3 denotes the symmetric group, distinct from the manifold S^3 used here), and Jakobs et al. [10] for dynamical continuum-limit studies.

Table 1: Plaquette enumeration on the S^3 Coulomb graph. V, E, c are vertex, edge, and component counts; $\beta_1 = E - V + c$ is the first Betti number; N_L is the number of primitive closed non-backtracking walks of length L (one representative per unoriented class).

$n_{\max}$	V	E	c	β_1	N_4	N_6	N_8
2	5	3	2	0	0	0	0
3	14	13	3	2	2	1	1
4	30	34	4	8	8	7	18

sector of its data. Section 7 states explicitly what is and is not claimed (Yang–Mills adjacency). Section 8 lists open questions, and Section 9 concludes.

2 Recap: The GeoVac Hopf Graph as a Lattice Gauge Structure

Paper 25 [18] specifies the combinatorial data in the form required here. We collect the definitions and fix notation; no new claim is made in this section.

The GeoVac graph at cutoff $n_{\max}$ has vertex set V indexed by quantum-number triples (n, l, m_l) with $n = 1, \ldots, n_{\max}$, $l = 0, \ldots, n-1$, $m_l = -l, \ldots, +l$, and $|V| = \sum_{n=1}^{n_{\max}} n^2$. Edges are of two kinds: $L_\pm$ angular ladders connecting $(n, l, m_l) \leftrightarrow (n, l, m_l \pm 1)$ at fixed (n, l), and $T_\pm$ radial ladders connecting $(n, l, m_l) \leftrightarrow (n', l, m_l)$ for $n' = n \pm 1$ (with selection-rule variants [13, 14]). Fix an orientation on each undirected edge; the signed incidence matrix $B \in \mathbb{Z}^{|V|\times|E|}$ has $B_{v,e} = +1$ if edge e starts at v, -1 if it ends at v, and 0 otherwise. The combinatorial Hodge Laplacians [5, 6] are

$$L_0 = BB^\top, \qquad L_1 = B^\top B, \tag{1}$$

the node and edge Laplacians respectively. The first Betti number

$$\beta_1 = |E| - |V| + c \tag{2}$$

(c = number of connected components) equals the kernel dimension of L_1 and counts the independent cycle classes.

The orienting convention for plaquettes differs here from the "elementary 4-cycle" convention of a hypercubic lattice. We take plaquettes to be the *primitive closed non-backtracking walks* of the graph, i.e. the Ihara cycles of Paper 29 [20]. A primitive closed non-backtracking walk on L edges is a sequence $e_1 e_2 \cdots e_L$ with $\mathrm{head}(e_i) = \mathrm{tail}(e_{i+1})$, no e_{i+1} equal to the reverse of e_i (including across the closure $e_L \to e_1$), and not a cyclic rotation of a shorter closed walk. This is the natural Wilson generalization on a non-hypercubic graph: primitive closed non-backtracking walks are exactly the irreducible "boundary of a disk" objects whose holonomies enter the Wilson action.

Table 1 collects the plaquette counts on the S^3 Coulomb graph at $n_{\max} = 2, 3, 4$. At $n_{\max} = 2$ the graph is a forest ($\beta_1 = 0$), so there are no plaquettes and the SU(2) Wilson theory is trivial ($Z(\beta) = 1$). At $n_{\max} = 3$ the theory is non-trivial for the first time: two primitive 4-cycles in the p-block, one primitive 6-cycle, one primitive 8-cycle. At $n_{\max} = 4$ the graph has $\beta_1 = 8$ and a richer plaquette structure.

The two primitive 4-cycles at $n_{\max} = 3$ both live in the p-block. With the canonical representative of each cycle chosen to begin at its lexicographically smallest oriented edge, they are

$$\begin{aligned} P_A &: (2,1,-1) \to (2,1,0) \to (3,1,0) \to (3,1,-1) \to (2,1,-1), \\ P_B &: (2,1,0) \to (2,1,1) \to (3,1,1) \to (3,1,0) \to (2,1,0). \end{aligned} \tag{3}$$

These share the undirected edge $\{(2,1,0),(3,1,0)\}$. The plaquette-sharing structure becomes important in Section 4: the leading-order character expansion treats plaquettes as independent, but the shared edge introduces higher-order corrections.

3 SU(2) Link Variables and the Wilson Action

3.1 Link variables

Each undirected edge $\{u,v\}$ with chosen orientation $u \to v$ carries a link variable $U_{u\to v} \in \mathrm{SU}(2)$. The reverse edge carries

$$U_{v\to u} = U_{u\to v}^{\dagger}, \tag{4}$$

so that each undirected edge stores one independent SU(2) element. An SU(2) matrix is parametrized by a unit quaternion (a,b,c,d) with $a^2+b^2+c^2+d^2=1$ as

$$U = \begin{pmatrix} a+id & c+ib \\ -c+ib & a-id \end{pmatrix}, \qquad \det U = 1, \quad UU^{\dagger} = I. \tag{5}$$

Equivalently, $U = \exp\bigl(i\theta\,\hat{n}\cdot\boldsymbol{\sigma}\bigr)$ where $\boldsymbol{\sigma} = (\sigma_x,\sigma_y,\sigma_z)$ are the Pauli matrices, $\hat{n}$ is a unit 3-vector, and θ is the half-rotation angle.

3.2 Plaquette holonomy

For a plaquette P traversed in the ordered sequence $e_1, e_2, \ldots, e_L$ (each e_i an oriented edge), the plaquette holonomy is the ordered product

$$U_P = U_{e_1} U_{e_2} \cdots U_{e_L}. \tag{6}$$

Since SU(2) is closed under matrix multiplication, $U_P \in \mathrm{SU}(2)$. The plaquette character

$$\chi(U_P) = \tfrac{1}{2}\mathrm{Re}\,\mathrm{tr}\,U_P = \cos\theta_P, \tag{7}$$

where θ_P is the half-angle of U_P in its SU(2) rotation parametrization, takes values in $[-1,+1]$ with $\chi(U_P) = +1$ iff $U_P = I$.

3.3 Wilson action

The Wilson action is

$$S_W[U] = \beta \sum_P \bigl(1 - \chi(U_P)\bigr) = \beta \sum_P \bigl(1 - \tfrac{1}{2}\mathrm{Re}\,\mathrm{tr}\,U_P\bigr), \tag{8}$$

with the sum ranging over the plaquettes enumerated as in Table 1. The gauge coupling is $\beta = 4/g^2$ in Wilson's convention: large β is weak coupling (continuum perturbation), small β is strong coupling (strong-lattice expansion). The action is non-negative and vanishes iff every plaquette holonomy is the identity (trivial vacuum).

3.4 Gauge transformation

A gauge transformation is a map $g : V \to \mathrm{SU}(2)$ acting on link variables as

$$U_{u\to v} \;\longmapsto\; g_u U_{u\to v} g_v^\dagger. \tag{9}$$

Under this action, the plaquette holonomy of any closed plaquette transforms by conjugation, $U_P \mapsto g_{v_0} U_P g_{v_0}^\dagger$, where v_0 is the base vertex. The trace $\operatorname{tr} U_P$ is therefore gauge-invariant, and so is the action S_W. Numerical verification on $n_{\max} = 3$ with random Haar link variables and random Haar gauge transformations gives $|S_W[U] - S_W[gUg^{-1}]| \leq 2.7 \times 10^{-15}$ (machine precision; worst case over 100 seeded Haar trials, and 1.8×10^{-15} in the recorded reference run); the committed tests assert $< 10^{-13}$ per trial — see `tests/test_su2_wilson_gauge.py`, class `TestGaugeInvariance` and `test_s3_coulomb_max_n3_endtoend`.

3.5 Partition function

The partition function is the Haar integral over $\mathrm{SU}(2)^{|E|}$,

$$Z(\beta) = \int \prod_e dU_e \; \exp\bigl(-S_W[U]\bigr), \tag{10}$$

with dU_e the normalized bi-invariant Haar measure on SU(2). This integral is finite-dimensional and well-defined for all $\beta \in \mathbb{R}$: no continuum renormalization is required. A direct implementation is via Monte Carlo sampling of link variables weighted by $\exp(-S_W)$; an analytical approach is via the SU(2) character expansion of Section 4.

4 Character Expansion on SU(2)

The SU(2) characters of the $(j+1)$-dimensional irreducible representations are

$$\chi_j(U) = \frac{\sin\bigl((j+1)\theta\bigr)}{\sin\theta}, \qquad U = e^{i\theta\,\hat{n}\cdot\boldsymbol{\sigma}}, \tag{11}$$

with the orthogonality $\int dU\,\chi_j(U)\,\overline{\chi_k(U)} = \delta_{jk}$. The Wilson Boltzmann factor expands as a Fegan–Menotti–Onofri series [4, 3, 2]

$$\exp\bigl(\beta \cdot \tfrac{1}{2}\mathrm{Re}\operatorname{tr} U\bigr) = \sum_{j=0}^{\infty} d_j(\beta)\,\chi_j(U), \qquad d_j(\beta) = \frac{2(j+1)}{\beta}\, I_{j+1}(\beta), \tag{12}$$

where I_n is the modified Bessel function of the first kind. At $\beta = 0$, $d_0 = 1$ and $d_j = 0$ for $j \geq 1$, so $Z(0) = 1$ (trivially, Haar measure is normalized). At $\beta \to \infty$, $d_j \to 1$ for all fixed j, and the Gaussian quadratic approximation around the trivial vacuum dominates (Section 5.2).

Coefficient table at $n_{\max} = 3$. For the two primitive 4-cycles of Eq. (3), the leading-order plaquette-independent partition function $Z_0(\beta) = e^{-\beta p}\bigl[d_0(\beta)\bigr]^p$ with $p = 2$ gives, together with the first excited coefficient $d_1(\beta) = (4/\beta) I_2(\beta)$,

The entries of Table 2 set the scale at which higher-rep corrections matter. For $\beta \leq 2$, $d_1/d_0 < 1$ and the leading-order approximation tracks Monte Carlo data (Section 5.3) to roughly 20%. For $\beta \geq 5$, higher-rep contributions compete with the trivial channel and the leading-order truncation loses accuracy.

Table 2: Character coefficients and leading-order partition function on the $n_{\max} = 3$ Hopf graph. The ratio d_1/d_0 exceeds unity at $\beta \approx 3$, so the plaquette-independence approximation loses accuracy for $\beta \gtrsim 5$. Numerical data: `debug/data/su2_wilson_gauge_results.json`.

β	$Z_0(\beta)$	$d_0(\beta)$	$d_1(\beta)$	d_1/d_0
0.1	8.21×10^{-1}	1.0013	0.0500	0.050
0.5	3.91×10^{-1}	1.0316	0.2552	0.247
1.0	1.73×10^{-1}	1.1303	0.5430	0.480
2.0	4.63×10^{-2}	1.5906	1.3779	0.866
5.0	4.30×10^{-3}	9.7343	14.0045	1.439
10.0	5.88×10^{-4}	534.20	912.61	1.708

5 Three Structural Results

5.1 Result 1: $U(1)$ reduction to Paper 25 is exact

Paper 25 defines the abelian $U(1)$ Wilson–Hodge action on the same graph by assigning a phase $\theta_e \in [0, 2\pi)$ to each oriented edge and setting

$$S_{U(1)}[\theta] = \beta \sum_{P} \big(1 - \cos\theta_P\big), \qquad \theta_P = \sum_{e \in P} \theta_e, \tag{13}$$

with reverse-edge phases negated and node-local gauge shifts $\theta_e \mapsto \theta_e + (\chi_{t(e)} - \chi_{s(e)})$.

Proposition 1 (Maximal-torus reduction)**.** *Let* $\mathrm{T} \subset \mathrm{SU}(2)$ *be the maximal torus,* $\mathrm{T} = \{\mathrm{diag}(e^{i\theta}, e^{-i\theta}) : \theta \in [0, 2\pi)\}$*. Restricting the link variables of the* $\mathrm{SU}(2)$ *Wilson action (8) to* T *yields the* $U(1)$ *Wilson action of Eq. (13):*

$$S_W\big[\{U_e = \mathrm{diag}(e^{i\theta_e}, e^{-i\theta_e})\}\big] = S_{U(1)}[\theta]. \tag{14}$$

Proof. A product of diagonal $\mathrm{SU}(2)$ matrices is diagonal; by induction on the plaquette length, $U_P = \mathrm{diag}(e^{i\theta_P}, e^{-i\theta_P})$ where $\theta_P = \sum_{e \in P} \theta_e$ (with reverse-edge phases counted negatively, matching the convention $U_{v \to u} = U_{u \to v}^\dagger$ which for diagonal links negates θ_e). Then $\frac{1}{2}\mathrm{Re}\,\mathrm{tr}\, U_P = \frac{1}{2}(e^{i\theta_P} + e^{-i\theta_P}) = \cos\theta_P$, and substitution into (8) yields (13). $\square$ $\square$

The proposition is verified numerically in `tests/test_su2_wilson_gauge.py`, class `TestU1Reduction`: for a triangle plaquette with random diagonal phases $\{\theta_e\} = \{0.3, 0.7, -0.5\}$ and $\beta = 1.7$, the two action evaluations agree to $|S_W - S_{U(1)}| < 10^{-12}$ (machine precision, 170 lines of test code). A fuller random-Haar-diagonal check at $\beta = 2$ on $n_{\max} = 3$ gives $|S_W - S_{U(1)}| = 0$ exactly.

Observation 1 (Paper 25 is the Cartan-subalgebra sector)**.** *Paper 25's abelian* $U(1)$ *Wilson–Hodge construction is structurally the Cartan-subalgebra projection of the present non-abelian* $\mathrm{SU}(2)$ *theory on the same graph. The question "is Paper 25 abelian because the underlying geometry forces it to be, or because abelian is one choice out of several?" has a definite answer: Paper 25 is the abelian sub-sector of a genuinely non-abelian lattice gauge theory on the same graph. This elevates Paper 25's* $U(1)$ *construction from a freestanding abelian model to the commutative-reduction sector of a natural non-abelian one.*

Remark 1 (higher-rank generalization)**.** *On a rank-r gauge group with maximal torus* $\mathrm{T} = U(1)^r$*, the analog of Proposition 1 produces r independent (but coupled, via a* $\det = 1$ *constraint when present)* $U(1)$ *sectors. The* $\mathrm{SU}(3)$ *case ($r = 2$) is computed in Sprint ST-SU3 (May 2026,* `debug/st_su3_wilson_me` *restricting Wilson* SU(3) *links on the Bargmann–Segal* S^5 *graph to the maximal torus* $T \cong U(1) \times U(1)$ *produces a coupled* $U(1)\times U(1)$ *Wilson action with character* $(1/3)\big(\cos a_P + \cos b_P + \cos(a_P + b_P)\big)$*, verified at machine precision over random Cartan trials at* $N_{max} = 2,3$ *(max* $|\Delta S| < 1.5 \times 10^{-11}$ *at* 7 765 *plaquettes).*

The character expansion of Section 4 reflects the reduction explicitly: the restriction of the SU(2) character χ_j to the maximal torus $U = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$ is

$$\chi_j\big(\mathrm{diag}(e^{i\theta}, e^{-i\theta})\big) = \frac{\sin\big((j+1)\theta\big)}{\sin\theta} = \sum_{k=0}^{j} e^{i(j-2k)\theta}, \tag{15}$$

a sum of $j+1$ $U(1)$ Fourier modes. The Fegan–Menotti–Onofri expansion (12) therefore restricts to the classical Jacobi–Anger expansion of $\exp(\beta\cos\theta)$ on the maximal torus—Paper 25's abelian expansion recovered term by term.

5.2 Result 2: the weak-coupling kinetic term completes $L_1 = B^\top B$ to the full Hodge-1 Laplacian

Paper 25 §II–III introduces the edge Laplacian $L_1 = B^\top B$ as the "discrete Hodge-1 Laplacian" whose kernel dimension equals β_1 and whose nonzero spectrum coincides with the nonzero spectrum of L_0 (by singular-value decomposition of B), and notes that on a bare 1-complex only the *exact* part $d_0 d_0^*$ of the manifold Hodge-1 operator $\Delta_1 = d_0 d_0^* + d_1^* d_1$ is available — the co-exact part requires faces. The present section identifies the *kinetic term* of the SU(2) Wilson action at weak coupling and shows it is exactly that missing co-exact (curl) part, built on the Ihara plaquettes as faces.

Write each link variable as an exponential of a $\mathfrak{su}(2)$-valued gauge field,

$$U_e = \exp\big(i\, A_e^a \sigma_a/2\big), \qquad a = 1,2,3, \tag{16}$$

with $A_e^a \in \mathbb{R}$ small (weak-coupling regime). To second order,

$$U_e = I + \tfrac{i}{2} A_e^a \sigma_a - \tfrac{1}{8} A_e^a A_e^b \sigma_a \sigma_b + O(A^3). \tag{17}$$

The plaquette holonomy U_P at second order in A is (using $\sigma_a\sigma_b = \delta_{ab} I + i\epsilon_{abc}\sigma_c$ to organize the Pauli products)

$$U_P = I + \tfrac{i}{2}\Big(\sum_{e\in P} A_e^a\Big)\sigma_a - \tfrac{1}{8}\Big(\sum_{e\in P} A_e^a\Big)\Big(\sum_{e'\in P} A_{e'}^a\Big) I + O(A^3) + (\text{commutators}), \tag{18}$$

where the "commutators" term is traceless and collects the non-abelian corrections $[A_e, A_{e'}]$ that appear at third order and higher. Taking $\frac{1}{2}\mathrm{Re}\,\mathrm{tr}$ and subtracting 1 gives

$$1 - \tfrac{1}{2}\mathrm{Re}\,\mathrm{tr}\, U_P = \tfrac{1}{8}\Big(\sum_{e\in P} A_e^a\Big)^2 + O(A^4) \qquad (\text{sum over } a = 1,2,3 \text{ implicit}). \tag{19}$$

The $O(A^3)$ term vanishes because $\mathrm{tr}\,\sigma_a = 0$. The sum $\sum_{e\in P} A_e^a$ is exactly the signed sum of the gauge field over the plaquette boundary, which is the action of the discrete exterior derivative $d_1 = B^\top$ (acting on 1-cochains; in fact the d_1 that maps 1-cochains to 2-cochains, but on a triangle-free 1-complex we can read it as a signed sum over plaquettes). Therefore

$$S_W[U] = \frac{\beta}{8}\sum_P \Big(\sum_{e\in P} A_e^a\Big)^2 + O(A^4) = \frac{\beta}{8}\, A^{a\top} M A^a + O(A^4), \tag{20}$$

where $M = B_2 B_2^\top$ is the plaquette-incidence bilinear form on the edge space, built from the signed edge–face incidence matrix B_2 whose columns are the (signed) indicator vectors of the Ihara plaquettes. In Hodge terms, $M = d_1^* d_1$ is the *co-exact (curl) part* of the Hodge-1 Laplacian of the 2-complex obtained by attaching the plaquettes as faces. It is *not* proportional to Paper 25's $L_1 = B^\top B = d_0 d_0^*$: the two quadratic forms have complementary supports. On the $n_{\max} = 3$ graph ($E = 13$, $\beta_1 = 2$) one computes $\operatorname{rank} M = 2 = \beta_1$, $\operatorname{rank} L_1 = 11 = E - \beta_1$, and $L_1 B_2 = 0$ exactly — the plaquette columns span $\ker L_1$ (the cycle space), which is precisely where the divergence form L_1 vanishes. Their sum $L_1 + M$ has full rank E and is the complete discrete Hodge-1 Laplacian of the plaquette-filled 2-complex (verified in `tests/test_su2_wilson_gauge.py`, class `TestWeakCouplingKinetic`). The structural statement is:

Proposition 2 (Wilson kinetic term = co-exact completion of L_1)**.** *The second-order expansion of the* SU(2) *Wilson action (8) around the trivial vacuum is*

$$S_W[U] = \tfrac{\beta}{8} \sum_a \langle A^a, B_2 B_2^\top A^a \rangle + O(A^4) + (\textit{non-abelian}) \tag{21}$$

where B_2 is the signed edge–plaquette incidence matrix (the discrete d_1-boundary data supplied by the Ihara plaquette enumeration). The kinetic quadratic form $B_2 B_2^\top$ is the co-exact (curl) part of the discrete Hodge-1 Laplacian of the plaquette-filled 2-complex; it is supported exactly on $\ker(L_1) = \ker(B^\top B)$*, the β_1-dimensional cycle space, and together with Paper 25's exact part it assembles the full Hodge-1 operator $\Delta_1^{disc} = B^\top B + B_2 B_2^\top$ (full rank at* $n_{\max} = 3$*). The* $\mathfrak{su}(2)$ *components A^a (*$a = 1, 2, 3$*) are kinetic-decoupled at quadratic order with the universal per-plaquette per-component coefficient* $1/8 = 1/(4N_c)|_{N_c=2}$*; non-abelian couplings first enter at order A^4.*

Provenance: MEASURED. The 1/8 coefficient and the quadratic form are verified numerically against exact matrix-exponential holonomies, both on a triangle plaquette and on the full $n_{\max} = 3$ graph (relative deviation 2.0×10^{-3} at $\epsilon = 0.01$ shrinking to 7.3×10^{-5} at $\epsilon = 0.005$, the $O(A^4)$ remainder), in `tests/test_su2_wilson_gauge.py::TestWeakCouplingKinetic` — the SU(2) mirror of the SU(3) 1/12 test in `tests/test_su3_wilson_s5.py`. An earlier version of this proposition asserted that the kinetic form "coincides with $L_1 = B^\top B$ up to a positive scalar multiple"; that assertion was wrong (the two forms have complementary supports, as the rank computation above shows) and is withdrawn — the correct statement is the completion statement, which matches the face-structure reading already recorded in Paper 25 §II.

Observation 2 (The Wilson kinetic sector completes the Hodge pair)**.** *At weak coupling the non-abelian Wilson theory contributes the co-exact half of the Hodge-1 pair: the free (Gaussian) sector is the curl form $B_2 B_2^\top$ on the cycle space, while Paper 25's $L_1 = B^\top B$ is the exact half, whose role is combinatorial (divergence / gradient sector) rather than plaquette-kinetic. The non-abelian self-couplings (cubic and quartic A-vertices) appear at order A^3 and A^4 in the Wilson action expansion and involve the structure constants of* $\mathfrak{su}(2)$*, namely ϵ^{abc}; they are parametrically suppressed at weak coupling. The QED role of the co-exact propagator $(B_2 B_2^\top)^+$ — recovering the continuum self-energy structural zero that L_1^+ alone cannot — is established in Paper 28 (see Paper 25 §II).*

Remark 2 (universal SU(N) kinetic coefficient)**.** *With generators $T^a = \lambda^a/2$ normalized to* $\operatorname{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$*, the second-order expansion of the Wilson action $S_W[U] = \beta \sum_P (1 - (1/N_c) \operatorname{Re} \operatorname{tr} U_P)$ on a general gauge group* SU(N_c) *produces a per-plaquette per-component coefficient of* $1/(4N_c)$*. The* SU(2) *value* 1/8 *derived above is the $N_c = 2$ specialization, verified numerically in* `tests/test_su2_wilson_gau`
The SU(3) *value* 1/12 *is verified independently in Sprint ST-SU3 (*`debug/st_su3_wilson_memo.md` *§ 4) symbolically and numerically on the Bargmann–Segal S^5 graph (*120 *degrees of freedom at $N_{max} = 2$,*

relative error $\sim O(\epsilon^2)$ consistent with $O(A^4)$ leading correction; frozen test **`tests/test_su3_wilson_s5.py::TestWe`** *The mechanism: more colors $\Rightarrow$ smaller per-component coefficient because the $(1/N_c)\,\mathrm{Re\,tr}$ normalization in the Wilson action divides by N_c while the trace-of-pair normalization holds $\mathrm{tr}(T^aT^a) = 1/2$ fixed. The decoupling of components at quadratic order is a universal $\mathrm{SU}(N_c)$ feature; non-abelian self-interactions enter at $O(A^4)$ via the structure constants f^{abc} in every case.*

5.3 Result 3: Plaquette counts and Wilson-loop data

Monte Carlo with Metropolis updates on the full $\mathrm{SU}(2)^{|E|}$ configuration space provides direct access to the Wilson-loop expectation value $\langle W_C\rangle = \left\langle \frac{1}{2}\mathrm{Re\,tr}\,U_C\right\rangle$. We run at $n_{\max} = 3$ (13 forward links), 2000 samples after 500 thermalization steps, seed = 12, for a Wilson loop coinciding with one of the two primitive 4-cycles in the p-block.

Table 3: Wilson-loop expectation on the $n_{\max} = 3$ Coulomb graph, loop = first p-block 4-cycle $[(2,1,-1) \to (2,1,0) \to (3,1,0) \to (3,1,-1)]$. MC: Metropolis sampling, 2000 samples, stderr reported; the seeded chain is bit-deterministic and the four rows are pinned by the frozen test `tests/test_su2_wilson_gauge.py::TestMonteCarloWilsonLoop` (slow-marked). Leading-order character expansion: $d_1(\beta)/d_0(\beta) = I_2(\beta)/I_1(\beta)$. Numerical data: `debug/data/su2_wilson_gauge_results.json`.

β	$\langle W\rangle_{\mathrm{MC}}$	$\pm\,$stderr	$\langle W\rangle_{\mathrm{char}}$
0.5	0.126	0.011	0.124
1.0	0.158	0.011	0.240
2.0	0.332	0.009	0.433
5.0	0.681	0.006	0.719

The two estimates agree at the 2–35% level (1.8%, 34%, 23%, 5.3% at $\beta = 0.5, 1, 2, 5$; loosest at intermediate β; corrected 2026-07-03 from an earlier "$\sim$ 20%" characterization that understated the $\beta = 1$ row). Discrepancy is structural: the leading-order character expansion treats plaquettes P_A and P_B of Eq. (3) as independent, which they are not (they share the undirected edge $\{(2,1,0),(3,1,0)\}$). Monte Carlo incorporates the shared-edge correlation. Both estimates agree that $\langle W\rangle$ is monotonic in β: strong coupling small β suppresses the trace, weak coupling large β drives $U_P \to I$ and $\langle W\rangle \to 1$.

Confinement diagnostic on a single-plaquette-class graph. Confinement in Wilson's sense is the statement that $\langle W_C\rangle$ obeys an *area law* $\langle W_C\rangle \sim e^{-\sigma\cdot\mathrm{Area}(C)}$ at strong coupling, crossing over to a *perimeter law* at weak coupling. The diagnostic requires a family of Wilson loops indexed by a geometric area. On the $n_{\max} = 3$ Coulomb graph, both primitive 4-cycles P_A and P_B have the same length (4) and the same "area" (they span 2×2 patches of the m_l-ladder / n-ladder lattice within the p-block), so a Wilson-loop family is not available. The Monte Carlo data do not (and cannot) show an area–perimeter crossover at this size. *This is a structural feature of the small graph, not a statement about the field theory.* To see confinement in the conventional sense, one needs a graph with a continuous family of Wilson-loop sizes—i.e., $n_{\max} \geq 5$ or finer discretizations of the continuum S^3. Section 8 lists this as an immediate follow-up.

Consistency with Paper 29's Ihara cycle counts. The primitive 4-cycle count at $n_{\max} = 3$ ($N_4 = 2$) and the higher-length counts ($N_6 = 1$, $N_8 = 1$) match the Möbius-inverted Ihara primitive-walk counts of Paper 29 [20] up to the orientation factor $N_{\text{Möbius}}(L) = 2\cdot N_{\text{unoriented}}(L)$ (each undi-

Table 4: Five action principles in the GeoVac framework. Each organizes a different sector of the graph data. The Wilson-action entry is the contribution of this paper.

Principle	Data sector	Extremum principle	Reference
Maximum entropy on packing	Nodes (candidate shell centers)	Maximize entropy under isotropy (Axiom 2)	Paper 0 [11]
Rayleigh–Ritz on graph Hamiltonian	Node wavefunctions (0-cochains)	Minimize $\langle\psi\vert H\vert\psi\rangle/\langle\psi\vert\psi\rangle$	Papers 1, 7 [13]
Wilson lattice gauge action	Edge link variables (1-cochains, SU(2)-valued)	Minimize $S_W[U] = \beta \sum_P (1 - \frac{1}{2}\mathrm{Re}\,\mathrm{tr}\,U_P)$	Paper 25 [18] $U(1)$; present paper SU(2)
Ihara zeta path integral	Primitive closed non-backtracking walks	$\log \zeta_G(s)^{-1} = \sum_{[C]} s^{\ell(C)}$ as weighted walk sum	Paper 29 [20]
Spectral action on Dirac spectrum	Spinor-bundle spectrum $(\vert\lambda_n\vert, g_n)$	$\mathrm{Tr}\, f\big(D/\Lambda\big)$ over modes	Paper 28 [19]

rected cycle gives clockwise and counterclockwise primitive walks; we keep one representative per unoriented class). Paper 29's plaquette-enumeration machinery is reused here without modification; the Ihara cycles of one paper are the Wilson plaquettes of the other.

6 Principle of Least Action: A Framework-Wide Synthesis

The Wilson action of Eq. (8) organizes the gauge-sector dynamics of the GeoVac graph as a variational principle on $\mathrm{SU}(2)^{|E|}$. This is the fifth independent action principle in the GeoVac framework, and it is useful to place it in context.

Over the course of the project, variational or action structures have emerged on five distinct data sectors of the Hopf graph. Each is tightly localized to one kind of structure and does not substitute for any of the others. We collect them in Table 4.

What each sector is about. The node sector carries the scalar matter content (the wavefunction amplitudes) and hosts two action principles: Paper 0's packing axioms as a max-entropy statement ("maximize entropy under isotropy"), which *builds* the graph, and the Rayleigh–Ritz principle on the built graph's Hamiltonian, which computes the graph's spectrum. The edge sector carries gauge content (Paper 25's $U(1)$ phases, this paper's SU(2) link variables) and hosts the Wilson action. The walk sector aggregates edge data into Ihara-cycle weights and hosts a discrete path integral (log-partition function = walk sum). The Dirac spectrum, finally, is the specific data set on which the spectral action of noncommutative-geometry type is computed; Paper 28's QED spectral sums are literally this.

The packing–Wilson duality. Two of the five principles stand out as structural duals. Paper 0's Axiom 2 (geometric indifference) is a max-entropy principle on node data: among all candidate discrete packings compatible with Axiom 1 (binary distinguishability), the entropy-maximizing one is selected. The Wilson action is a min-action principle on edge data: among all SU(2) link configurations, those minimizing plaquette curvature are selected in the continuum ($\beta \to \infty$) limit. The two principles live on complementary cochain degrees (0-form vs 1-form) and extremize in opposite directions (max vs min), but they are structurally compatible: Paper 0's max-entropy on nodes fixes *which* graph the Wilson action lives on, and the Wilson action fixes *how* gauge dynamics runs on it. Nothing in the mathematics forbids a joint variational principle on the full (node, edge) data; we note this for future work but do not pursue it here.

What this section does and does not claim. The observation is descriptive: the GeoVac framework, as it currently stands, presents five independent variational structures, each localized to a different sector of the graph data. We do not claim that these five can be reduced to a single grand principle, nor that one is more fundamental than the others. Each is a tool matched to a data type. The utility of naming them together is that it makes the project's full toolkit visible in one place: the framework is not reducible to "diagonalize a graph Laplacian," because its data has more kinds of structure than nodes alone.

7 Yang–Mills Adjacency: What Is and Is Not Claimed

The present paper inherits the language of Wilson lattice gauge theory. The rhetoric rule of CLAUDE.md §1.5 requires that we be explicit about scope. We enumerate what is done and what is not.

What this paper establishes. A gauge-invariant, finite, Haar-normalizable non-abelian lattice gauge theory on the finite compact GeoVac Hopf graph at any $n_{\max}$, with (i) link variables in SU(2), (ii) plaquettes as primitive closed non-backtracking walks, (iii) Wilson action and partition function well-defined without renormalization, (iv) exact $U(1)$ reduction to Paper 25 in the maximal-torus limit, (v) the weak-coupling kinetic term identified as the co-exact (curl) completion of $L_1 = B^\top B$, with universal coefficient $1/8 = 1/(4N_c)$. All 36 tests in `tests/test_su2_wilson_gauge.py` pass, including the numerical $U(1)$-reduction test, end-to-end gauge invariance at machine precision (asserted $< 10^{-13}$), the kinetic quadratic-form tests, and the seeded Monte Carlo Wilson-loop table.

What this paper does NOT establish.

- *Yang–Mills mass gap on* $\mathbb{R}^4$. The Clay Millennium Problem requires a continuum-field-theory construction of pure Yang–Mills on 4D Minkowski or Euclidean space, proof of a positive mass gap, and proof that no interacting scale-invariant continuum limit exists. Our construction lives on a fixed finite compact graph; there is no continuum limit of $\mathbb{R}^4$ inside it.

- *Confinement in the conventional sense.* As noted in Section 5.3, the $n_{\max} = 3$ graph has a single plaquette-length class, so the area/perimeter-law diagnostic is degenerate. Finite-size graphs with multiple plaquette classes (e.g., $n_{\max} \geq 5$) would be needed to even state the diagnostic.

- *Lorentzian signature or dynamics.* This is a Euclidean lattice gauge theory on a spatial slice: no time direction, no Wick rotation, no S-matrix, no scattering. Dynamics in the usual sense is absent.

- *Renormalization or continuum limit.* The bare Wilson action is the whole story on a finite graph. Whether the $n_{\max} \to \infty$ limit recovers continuum SU(2) Yang–Mills on the unit S^3, and in what sense, is an open question we do not address here.
- *No new derivation of α.* Paper 2's combination rule $K = \pi(B + F - \Delta)$ is not refined by the present construction. Paper 25 §IV read the three ingredients as Casimir-matter trace / Fock-Dirichlet gauge zeta / Dirac-boundary count; the non-abelian SU(2) Wilson structure does not produce a new K-like object or a non-abelian "fine-structure constant" coupling, and we make no claim in that direction.

Relationship to the Clay problem. The mathematical vocabulary is shared (lattice gauge theory, non-abelian link variables, plaquette action, Wilson loops), but the setting differs categorically. The Clay problem is about continuum field theory on a non-compact 4D spacetime; the present construction is about discrete gauge theory on a compact 3D-graph spatial slice. We use Wilson's formalism because it is the correct tool for non-abelian lattice gauge theories on finite graphs; we do not imply that the construction transports to $\mathbb{R}^4$.

8 Open Questions

8.1 Matter coupling: Dirac fermions on the Hopf graph

The GeoVac framework's Tier-2 relativistic pipeline [14] builds spin-ful composed qubit Hamiltonians in the (κ, m_j) Dirac basis on S^3. These are matter sectors on the same graph whose gauge sector we have built here. Coupling the two—Dirac fermion matter on nodes, SU(2) Wilson gauge on edges—would be the first matter+gauge lattice theory in the GeoVac project. The construction is standard in principle (covariant hopping amplitudes $\bar{\psi}_i U_{ij} \psi_j$), but on a triangle-free simplicial 1-complex with Dirac spinors it would require the Szmytkowski angular labels of Paper 14's Tier 2 to be reconciled with the Wilson link-variable orientation convention. A clean demonstrator would be: does the spinor spectrum feel the non-trivial SU(2) topology (i.e., is the Dirac operator sensitive to the $\beta_1 = 2$ harmonic 1-forms at $n_{\max} = 3$ when the gauge field is non-trivially topological)?

8.2 Larger graphs and the confinement diagnostic

At $n_{\max} \geq 5$, the Coulomb graph acquires multiple plaquette-length classes, making a Wilson-loop family indexed by area available for the first time. The natural question: on the $n_{\max} = 5, 6$ Coulomb graphs, does the Wilson loop exhibit area-law behavior at small β and perimeter-law behavior at large β? Running this is computationally straightforward (the Metropolis update scales as $|E|^2$ per sweep; $n_{\max} = 5$ has $|E| \sim 90$, manageable). The result would distinguish a confining non-abelian theory from a deconfined one on the GeoVac graph; Paper 25 §VII's broader questions about lattice-gauge structure would receive an immediate sharpening.

8.3 SU(2) analog of the Ihara zeta

Paper 29 built the (scalar-valued) Ihara zeta of the GeoVac Hopf graphs from primitive closed non-backtracking walk counts. A natural non-abelian generalization would weight each primitive walk $[C]$ by a character $\chi_j(U_{[C]})$ of its SU(2) holonomy, producing a character-valued zeta function. Whether the resulting object has a well-defined Ihara–Bass-type determinantal expression, or whether it satisfies a graph-Riemann Hypothesis analog, is speculative. We record the question without pursuing it.

8.4 SU(2) analog of Paper 2's $K = \pi(B + F - \Delta)$

Paper 25 §VII.A suggested that a non-abelian analog of the Paper 2 combination rule might exist on the S^3 graph. The SU(2) Wilson structure of the present paper does not by itself produce one; the Wilson action is coupling-insensitive in a way that Paper 2's K is not. A more promising route (if any) would couple the SU(2) Wilson gauge to Dirac-spinor matter (first open question above) and examine the combined spectral-gauge invariants. We do not claim such a route will succeed; we record it as a natural follow-up within the Paper-2 observation-tier program.

8.5 Plaquettes as the transverse photon sector

The plaquettes enumerated in this paper define a boundary operator $d_1 : C^1 \to C^2$ whose transpose gives the co-exact Laplacian $L_1^{\mathrm{co}} = d_1 d_1^\top$ on edges. This is the *transverse* (physical photon) sector of the graph Hodge decomposition, complementing the exact (longitudinal) sector $L_1^{\mathrm{ex}} = B^\top B$ used in Paper 25 and the graph-native QED of Paper 28.

Paper 28's pendant-edge theorem shows that the scalar propagator $G = (B^\top B)^+$ breaks the continuum self-energy structural zero: $\Sigma_{\mathrm{scalar}}(\mathrm{GS}) = 2(n_{\max} - 1)/n_{\max}$. Replacing G with the transverse propagator $G_T = (d_1 d_1^\top)^+$ recovers the structural zero *exactly*: $\Sigma_T(\mathrm{GS}) = 0$ at $n_{\max} = 3, 4$ (verified to machine precision). The mechanism is topological: the ground-state node is a pendant vertex whose unique edge cannot participate in any closed walk, so $d_1[e_0, :] = 0$ and hence $G_T[e_0, :] = 0$ identically.

This gives the plaquettes of this paper a concrete QED role beyond their original gauge-theory motivation: they define the physical photon sector that recovers continuum selection rules inaccessible from the scalar (longitudinal) propagator alone. One of the four "vector-photon-required" selection rules identified in Paper 28's census is now "plaquette-recoverable," refining the three-tier partition to four tiers.

8.6 Continuum limit

The continuum limit of lattice gauge theory on refined Alon–Boppana sequences of graphs converging to a target manifold is a technical question that has been addressed in specific cases (hypercubic lattices converging to $\mathbb{R}^4$, spherical lattices converging to S^3). Whether the $n_{\max} \to \infty$ sequence of GeoVac graphs admits a well-posed continuum limit, and whether that limit is SU(2) Yang–Mills on the unit S^3 in the Hodge–de Rham sense, is open. The relevant spectral data is available in `geovac/hopf_bundle.py`; a careful analysis has not been done.

8.7 SU(3) Wilson sibling on S^5

Paper 25 § VII.A also flagged as open whether a non-abelian extension of the GeoVac Hopf $U(1)$ Wilson–Hodge structure transfers to the Bargmann–Segal S^5 graph (Paper 24). Sprint 5 Track S5 (`debug/s5_gauge_structure_memo.md`) returned a clean negative for the *adjacency-preserving* SU(3) action on the $(N, 0)$ representation tower: shell-shifting transitions are Clebsch–Gordan intertwiners between distinct SU(3) irreps, not group elements, so an SU(3) action on the $(N, 0)$ tower fails the Wilson "fixed group on every link" requirement.

Sprint ST-SU3 (May 2026, `debug/st_su3_wilson_memo.md`, with module `geovac/su3_wilson_s5.py` and tests `tests/test_su3_wilson_s5.py` 49/49 pass, two slow-marked) tested a structurally different question: *Wilson lattice gauge theory* with link variables $U_e \in \mathrm{SU}(3)$ as external 3×3 matrices in the fundamental representation, independent of the shell index N. The construction parallels the present paper's SU(2) on S^3. Three structural results mirror those of the present paper:

- **Cartan reduction.** Restricting Wilson SU(3) links to the maximal torus $T \cong U(1) \times U(1)$ produces a coupled $U(1)\times U(1)$ Wilson action with character $(1/3)\big(\cos a_P + \cos b_P + \cos(a_P + b_P)\big)$ (the third phase forced by $\det = 1$). Verified at machine precision over random Cartan trials at $N_{\max} = 2, 3$ (max $|\Delta S| < 1.5 \times 10^{-11}$ at 7 765 plaquettes). The rank-2 generalization of Proposition 1 per Remark 1.
- **Kinetic coefficient.** The weak-coupling expansion gives $1/12 = 1/(4 \cdot 3)$ per plaquette per generator component, the $N_c = 3$ specialization of Remark 2's universal $1/(4N_c)$ formula. Eight $\mathfrak{su}(3)$ components are kinetic-decoupled at quadratic order; non-abelian self-couplings enter at $O(A^4)$ via the structure constants f^{abc}.
- **Plaquette tabulation.** V, E, β_1 at $N_{\max} = 2, 3, 4$ are $(10, 15, 6)$, $(20, 42, 23)$, $(35, 90, 56)$, with primitive 4-cycle counts $N_4 = 15, 88, 272$ — roughly an order of magnitude denser than the S^3 graph at matched cutoff. Wilson loops at $N_{\max} = 2$ with local SU(3) updates give $\langle W \rangle = 0.05, 0.10, 0.26, 0.65, 0.89$ at $\beta = 0.5, 1, 2, 5, 10$, systematically lower than the present paper's SU(2) values at small β (consistent with the standard N_c-color randomization of the trace).

The Sprint 5 Track S5 Clebsch–Gordan obstruction is *bypassed* at the gauge level (Wilson links live in the fundamental SU(3) rep, not in the $(N, 0)$ shell rep) but *rediscovered* at the matter-coupling level: coupling Wilson SU(3) to natural shell matter (in the $(N, 0)$ irreps of growing dimension) requires a CG intertwiner from $\mathbf{3} \otimes (N, 0)$ to $(N + 1, 0)$ that is exactly the Sprint 5 obstruction. Wilson SU(3) on Bargmann is therefore a self-consistent *pure-gauge* theory; the natural matter sector cannot be coupled in the strict Wilson sense.

The structural reading: $S^3 = \mathrm{SU}(2)$ makes the present paper's case special in a way that does not transfer to the SU(3) on S^5 analogue. On S^3, matter (spin-1/2 fermions, Paper 14 Tier 2) lives in the same representation as the link variables, so the gauge–matter coupling is natural; on S^5, the matter representations $(N, 0)$ have dimensions that grow with N, so a uniform-link-rep coupling is impossible. This sharpens Paper 24's Coulomb/HO asymmetry from three layers (spectrum-computing L_0, calibration π, Wilson gauge with matter coupling) plus the universal Wilson–Hodge vocabulary, to *four layers*: Wilson gauge alone is universal; only Wilson gauge *with natural matter coupling* is Coulomb-specific. Sprint ST-SU3 contributes the third layer on the gauge-with-matter side. The reading of Paper 24 § V is updated accordingly.

9 Conclusion

Sprint 4 Track RH-Q constructed SU(2) Wilson lattice gauge theory on the GeoVac Hopf graph (module `geovac/su2_wilson_gauge.py`, 36 tests passing, 2026-07-04 count). The present paper formalizes that construction and places it in context relative to Paper 25's abelian $U(1)$ predecessor.

Three structural results summarize the content: (1) the maximal-torus reduction of the SU(2) Wilson action recovers Paper 25's $U(1)$ Wilson–Hodge action exactly, so Paper 25 is the Cartan-subalgebra sector of a non-abelian theory; (2) the weak-coupling kinetic term of the non-abelian Wilson action is the co-exact (curl) quadratic form on the Ihara plaquettes — coefficient $1/8 = 1/(4N_c)$ per plaquette per $\mathfrak{su}(2)$ component — which completes the edge Laplacian $L_1 = B^\top B$ that Paper 25 introduced to the full discrete Hodge-1 Laplacian, with non-abelian self-interactions entering only at quartic order in the gauge field; (3) plaquette enumeration on the S^3 Coulomb graph matches Paper 29's Ihara cycle counts, and direct Monte Carlo on $n_{\max} = 3$ gives monotonic β-dependence of the Wilson loop expectation (confirming the character-expansion leading order at the $\sim 20\%$ level, as expected from the shared-edge correction).

We close with a framework observation. The $S^3 = \mathrm{SU}(2)$ coincidence is structurally overdetermined: S^3 appears in the framework as Fock's projection of the Coulomb problem (Paper 7), as the underlying manifold of the (n, l, m_l) Hopf graph (Paper 14), as the spin-1/2 carrier of the Dirac sector (Paper 14 Tier-2), and now as the gauge group of the non-abelian Wilson construction. Four independent structural roles on the same 3-manifold, each needing only the simplest compact simply-connected non-abelian Lie group of rank 1. We do not read this as evidence that S^3 is fundamental in any ontological sense (CLAUDE.md §1.5 rhetoric); we do observe that S^3 is, in a specific technical sense, the minimal object that supports all four roles simultaneously, and we record the convergence as a fact.

The project now contains five action principles—Paper 0's max-entropy on nodes, Rayleigh–Ritz on the graph Hamiltonian, Paper 25's and this paper's Wilson action on edges, Paper 29's Ihara zeta path integral on walks, and Paper 28's spectral action on the Dirac spectrum—each matched to a distinct sector of the graph data. None reduces to any of the others. The Wilson action has, with this paper, a genuine non-abelian sibling to its $U(1)$ predecessor; the rest of the table is left as is.

References

[1] K. G. Wilson, "Confinement of quarks," *Phys. Rev. D* **10**, 2445–2459 (1974).

[2] M. Creutz, *Quarks, Gluons and Lattices*, Cambridge University Press, Cambridge (1983).

[3] P. Menotti and E. Onofri, "The action of SU(N) lattice gauge theory in terms of the heat kernel on the group manifold," *Nucl. Phys. B* **190**, 288–300 (1981).

[4] H. D. Fegan, "The heat equation on a compact Lie group," *Trans. Amer. Math. Soc.* **246**, 339–357 (1978).

[5] L.-H. Lim, "Hodge Laplacians on graphs," *SIAM Review* **62**, 685–715 (2020).

[6] M. T. Schaub, A. R. Benson, P. Horn, G. Lippner, and A. Jadbabaie, "Random walks on simplicial complexes and the normalized Hodge 1-Laplacian," *SIAM Review* **62**, 353–391 (2020).

[7] M. Marcolli and W. D. van Suijlekom, "Gauge networks in noncommutative geometry," *J. Geom. Phys.* **75**, 71–91 (2014), arXiv:1301.3480.

[8] C. I. Perez-Sanchez, "Comment on 'Gauge networks in noncommutative geometry',", arXiv:2508.17338 (2025).

[9] M. Garofalo, T. Hartung, T. Jakobs, K. Jansen, J. Ostmeyer, D. Rolfes, S. Romiti, and C. Urbach, "Testing the SU(2) lattice Hamiltonian built from S_3 partitionings," arXiv:2311.15926 (2023).

[10] T. Jakobs, M. Garofalo, T. Hartung, K. Jansen, J. Ostmeyer, S. Romiti, and C. Urbach, "Dynamics in Hamiltonian lattice gauge theory: Approaching the continuum limit with partitionings of SU(2)," arXiv:2503.03397 (2025).

[11] GeoVac Project, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," Paper 0 (2025).

[12] GeoVac Project, "The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration," Paper 2 (2025).

[13] GeoVac Project, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," Paper 7 (2025).

[14] GeoVac Project, "Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory," Paper 14 (2025).

[15] GeoVac Project, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," Paper 18 (2026).

[16] GeoVac Project, "Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems," Paper 22 (2026).

[17] GeoVac Project, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," Paper 24 (2026).

[18] GeoVac Project, "The Hopf Graph as a Lattice Gauge Structure: A Framework Observation," Paper 25 (2026).

[19] GeoVac Project, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," Paper 28 (2026).

[20] GeoVac Project, "Finite-Size Graph-RH for the GeoVac Hopf Graphs: Ihara Zeta, Bound-Crossing, and a Scope Boundary on Selberg-on-Hydrogen," Paper 29 (2026).

[21] GeoVac Project, "Track RH-Q: SU(2) Wilson Lattice Gauge Theory on the $S^3 = \mathrm{SU}(2)$ GeoVac Hopf Graph," `debug/su2_wilson_gauge_memo.md`, April 2026.